

FYS-STK August 18, 2026



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$$C(e) = \frac{1}{n} \sum_{i=0}^{n-1} (y_i - \tilde{y}_i)^2$$

(MSE)

$$y_i = f(x_i) + \varepsilon_i \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2)$$

Model $\tilde{y}_i \approx f(x_i)$ $\mu=0$ $\sigma^2=1$

polynomial $\tilde{y}_i = \sum_{j=0}^{p-1} x_i^j \Theta_j$

$$X = \begin{bmatrix} x_{00} & x_{01} & - & - & x_{0p-1} \\ x_{10} & - & - & - & x_{1p-1} \\ \vdots & & & & \\ x_{n-10} & - & - & - & x_{n-1p-1} \end{bmatrix}$$

$$C(\theta) = \frac{1}{n} \sum_{i=0}^{n-1} (y_i - \underbrace{\sum_j x_{ij} \theta_j}_{\hat{y}_i})^2$$

$$\frac{\partial C(\theta)}{\partial \theta_j} = 0$$

$$= -\frac{2}{n} \sum_{i=0}^{n-1} x_{ij} (y_i - \sum_k \theta_k x_{ik})$$

$$= 0$$

$$(p=2 \quad n=2)$$

$$\frac{\partial C}{\partial \theta_j} \rightarrow \vec{\nabla}_{\theta} C(\theta) = 0$$

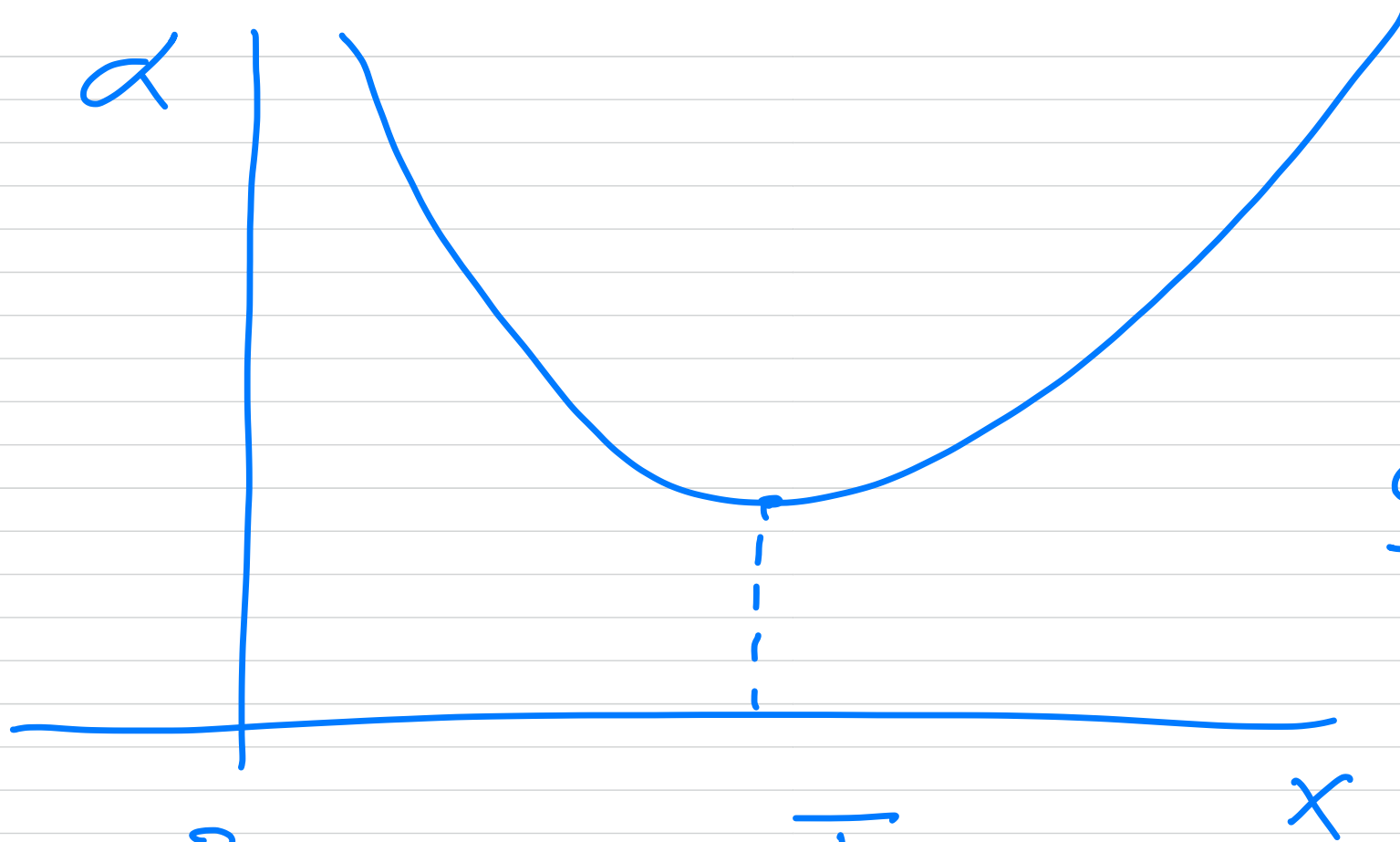
$$= -\frac{2}{n} X^T (\vec{y} - X \vec{\theta})$$

$$X^T X \hat{\beta} = X^T y \Rightarrow$$

$$\hat{\beta}_{OLS} = (X^T X)^{-1} X^T y$$

$$\hat{y}_{opt} = X \hat{\beta}_{OLS}$$

$$\sigma_{\beta}^2 C = \frac{2}{n} X^T X$$

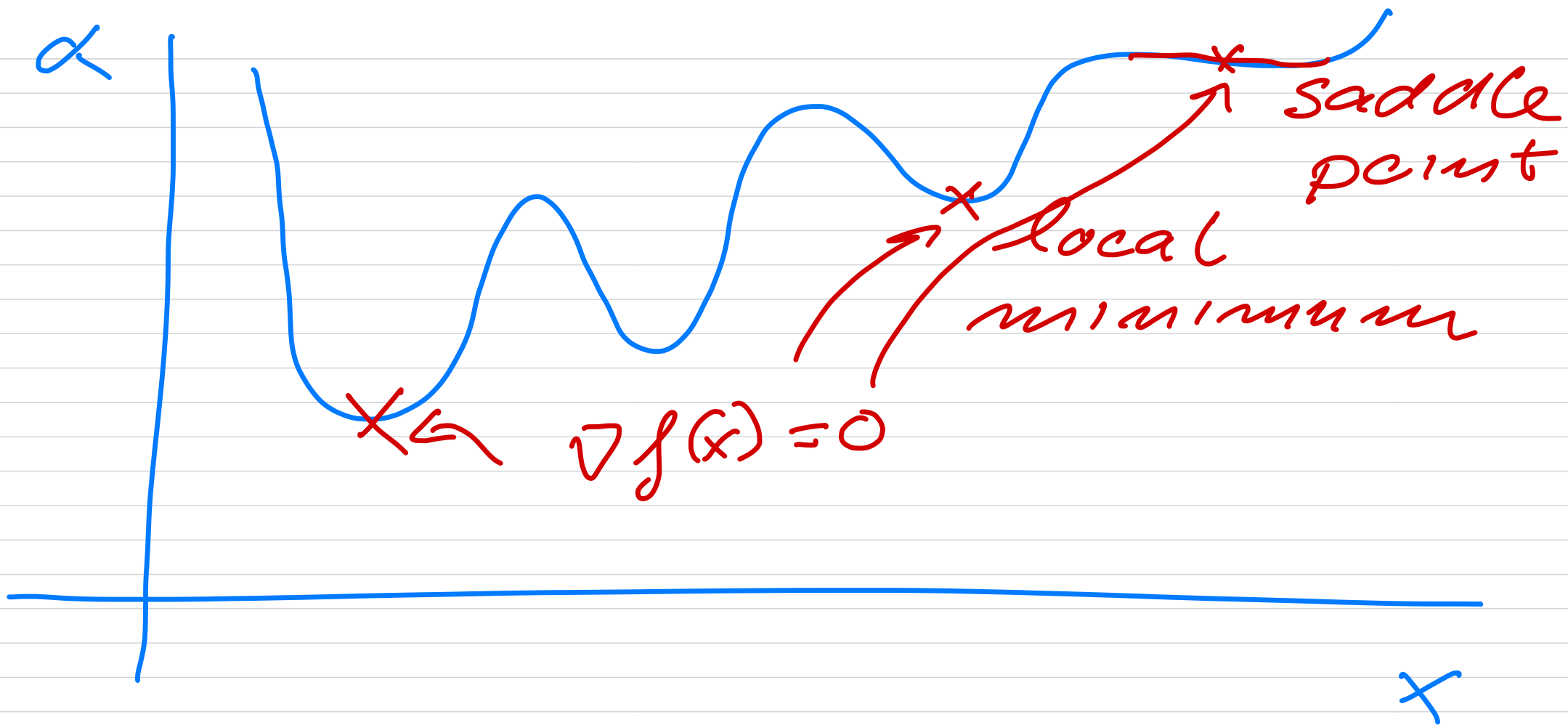


$$\alpha = f(x)$$

$$\frac{d^2 f}{dx^2} \geq 0$$

$$\nabla^2_{\Theta} = \frac{2}{n} X X^T$$

Hessian with
eigenvalues
 $\lambda_i \geq 0$



1.18 - 1.20 SVD

$$A \in \mathbb{R}^{n \times p}$$

$$A = U \Sigma V^T$$

Diagram illustrating the dimensions of the matrices in the SVD decomposition:

- $U \in \mathbb{R}^{n \times m}$ (indicated by an arrow from U to $\mathbb{R}^{n \times m}$)
- $\Sigma \in \mathbb{R}^{n \times p}$ (indicated by an arrow from Σ to $\mathbb{R}^{n \times p}$)
- $V^T \in \mathbb{R}^{p \times p}$ (indicated by an arrow from V^T to $\mathbb{R}^{p \times p}$)

$$U U^T = U^T U = \underline{\underline{1}}$$

$$V V^T = V^T V = \underline{\underline{1}}$$

$$\Sigma = \begin{bmatrix} \lambda_0 & & & 0 \\ & \lambda_1 & & \\ & & \ddots & \\ & & & \lambda_{p-1} \\ & & & & 0 \end{bmatrix}$$

$$\lambda_0 > \lambda_1 > \lambda_2 > \dots > \lambda_{p-1} > 0$$

$$\Sigma = \begin{bmatrix} 2 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$$

$$\lambda_0 = 2 \quad \lambda_1 = 1$$

$$\Sigma^T \Sigma = \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\Sigma \Sigma^T = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$U = \begin{bmatrix} \vec{u}_0 & \vec{u}_1 & \dots & \vec{u}_{n-1} \end{bmatrix}$$

$$u_i^T u_j = \delta_{ij}$$

$$V = \begin{bmatrix} \vec{v}_0 & \vec{v}_1 & \dots & \vec{v}_{p-1} \end{bmatrix}$$

$$\underbrace{X^T X}_{p \times p} = \underbrace{V \Sigma^T}_{p \times n} \underbrace{U^T U}_{\underset{n \times n}{\mathbb{I}}} \underbrace{\Sigma V^T}_{n \times p}$$

$$\begin{aligned}
 V \Sigma^T \Sigma &= V \left(\Sigma^T \Sigma \right) V^T \\
 &= \left(\Sigma^T \Sigma \right) V
 \end{aligned}$$

$$[\overline{X^T X} = (\Sigma^T \Sigma) v v^T] \cdot v$$

$$v^T v = \underline{\underline{1}}$$

$$\overline{X^T X} v = (\Sigma^T \Sigma) v$$

$$\boxed{\overline{X^T X} v_i = \sigma_i^2 v_i}$$

eigenvalue problem
 $\sigma_i^2 \geq 0$

$$\underbrace{(X^T X)^{-1}} X^T y = \hat{\theta}_{OLS}$$

if singular, then no
OLS solution.

$$\lambda \gg 0$$

$X^T X + \lambda \mathbb{1}$, then we
can invert $X^T X + \lambda \mathbb{I}$

$$\hat{E}_{\text{Ridge}} = \underbrace{(X^T X + \lambda \mathbb{1})}_{p \times p}^{-1} \underbrace{X^T y}_{p \times 1}$$

$$\begin{aligned} C_{\text{Ridge}} &= \frac{1}{n} \sum_{i=0}^{n-1} (y_i - \sum_j x_{ij} \theta_j)^2 \\ &\quad + \lambda \sum_{j=0}^{p-1} \theta_j^2 \\ &= \frac{1}{n} \sum_{i=0}^{n-1} (y_i - \hat{y}_i)^2 + \lambda \|\theta\|_2^2 \end{aligned}$$

$$\vec{\nabla}_{\mathbf{E}} C_{\text{Ridge}} = - \frac{2}{n} \mathbf{X}^T (\mathbf{y} - \mathbf{X}\mathbf{E})$$

$$+ 2\lambda \sum_j \mathbf{E}_j = 0$$

$$\mathbf{X}^T \mathbf{X} \mathbf{E} + \underbrace{\lambda n}_{\lambda} \mathbf{E} = \mathbf{X}^T \mathbf{y}$$

$$\hat{\mathbf{E}}_{\text{Ridge}} = (\mathbf{X}^T \mathbf{X} + \lambda \mathbf{1})^{-1} \mathbf{X}^T \mathbf{y}$$

$$C_{\text{lasso}} = \frac{1}{n} \|y - X\theta\|_2^2 + \lambda \|\theta\|_1$$

$$\|x\|_1 = \sum_j |x_j|$$